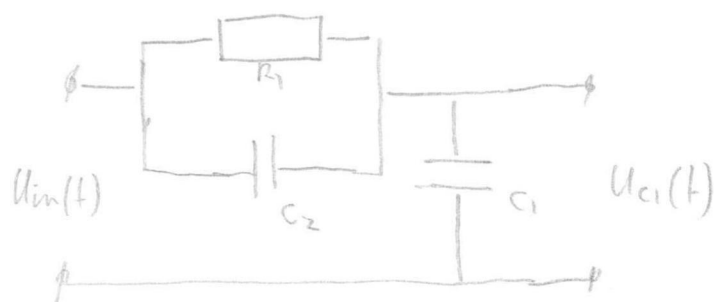


opgave H2.9.6b

HTJ



$$I_{C1(t)} = C \cdot u'_{C1(t)} \Leftrightarrow$$

$$u'_{C1(t)} = \frac{1}{C} \cdot I_{C1(t)}$$

stel $H(s) = \frac{u_{C1}(s)}{u_{in}(s)}$ op.

(Deze opgave kan op veel manieren worden uitgewerkt. Onderstaande uitwerking is slechts één van de vele mogelijke aanpakken.)

* $u_{in}(t) = u_{C2}(t) + u_{C1}(t) \Leftrightarrow u'_{in}(t) = u'_{C2}(t) + u'_{C1}(t) = \frac{1}{C_2} \cdot I_{C2(t)} + u'_{C1}(t) \quad (1)$

* Na $I_{C2(t)}$ uitschakken in $u_{in}(t)$ en $u_{C1}(t)$:

$$\begin{aligned} I_{C2(t)} &= I_{C1(t)} - I_{R1(t)} = C_1 \cdot u'_{C1(t)} - \frac{1}{R_1} \cdot u_{R1}(t) = \\ &= C_1 \cdot u'_{C1(t)} - \frac{1}{R_1} \cdot (u_{in}(t) - u_{C1}(t)) = \\ &= C_1 \cdot u'_{C1(t)} - \frac{1}{R_1} \cdot u_{in}(t) + \frac{1}{R_1} \cdot u_{C1}(t) \end{aligned} \quad (2)$$

* (1), (2): $u'_{in}(t) = \frac{1}{C_2} \left(C_1 \cdot u'_{C1}(t) - \frac{1}{R_1} \cdot u_{in}(t) + \frac{1}{R_1} \cdot u_{C1}(t) \right) + u'_{C1}(t) \Leftrightarrow$

$$R_1 \cdot C_2 \cdot u'_{in}(t) = R_1 \cdot C_1 \cdot u'_{C1}(t) - u_{in}(t) + u_{C1}(t) + R_1 \cdot C_2 \cdot u'_{C1}(t)$$

* Na Laplace-transformatie: $R_1 \cdot C_2 \cdot s \cdot u_{in}(s) = R_1 \cdot C_1 \cdot s \cdot u_{C1}(s) - u_{in}(s) + u_{C1}(s) + R_1 \cdot C_2 \cdot s \cdot u_{C1}(s)$

$$\Leftrightarrow (R_1 \cdot C_2 \cdot s + 1) \cdot u_{in}(s) = (R_1 \cdot C_1 \cdot s + 1 + R_1 \cdot C_2 \cdot s) \cdot u_{C1}(s)$$

$$\Leftrightarrow \frac{u_{C1}(s)}{u_{in}(s)} = \frac{R_1 \cdot C_2 \cdot s + 1}{R_1 \cdot C_1 \cdot s + 1 + R_1 \cdot C_2 \cdot s} = \frac{R_1 \cdot C_2 \cdot s + 1}{(C_1 + C_2) \cdot R_1 \cdot s + 1}$$